

Step :1 Install SonarQube Server on WSL

- Requirements for this are
 - Must have JDK 17 and above in wsl
 - sudo apt update
 - sudo apt install openjdk-17-y
 - java -version

```
nikunj@DESKTOP-M4S6FPT:~$ java -version
openjdk version "17.0.15" 2025-04-15
OpenJDK Runtime Environment (build 17.0.15+6-Ubuntu-0ubuntu124.04)
OpenJDK 64-Bit Server VM (build 17.0.15+6-Ubuntu-0ubuntu124.04, mixed mode, sharing)
nikunj@DESKTOP-M4S6FPT:~$ |
```

Step:2 Install PostgreSQL

- sudo apt install postgresql postgresql-contrib -y

Now Create SonarQube DB and User:

- sudo -u postgres psql
- CREATE USER sonar WITH PASSWORD 'sonar'
- CREATE DATABASE sonarqube OWNER sonar;

```
postgres=# CREATE USER sonar WITH PASSWORD 'sonar'
CREATE DATABASE sonarqube OWNER sonar
postgres=# |
```

- \q

```
postgres=# \q|
```

Step:3 Install required Dependencies

- sudo apt install unzip wget gnupg2 -y

```
nikunj@DESKTOP-M4S6FPT:~$ sudo apt install unzip wget gnupg2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
unzip is already the newest version (6.0-28ubuntu4.1).
wget is already the newest version (1.21.4-1ubuntu4.1).
gnupg2 is already the newest version (2.4.4-2ubuntu17.3).
The following package was automatically installed and is no longer required:
  libltdl7
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 8 not upgraded.
nikunj@DESKTOP-M4S6FPT:~$ |
```

- sudo wget <https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-10.5.1.90531.zip>
- sudo unzip sonarqube-10.5.1.90531.zip
- sudo mv sonarqube-10.5.1.90531.zip sonarqube

- `sudo chown -R $USER:$USER sonarqube`
- `nano /opt/sonarcube/conf/sonar.properties`
(if opt folder is not available create using command “`mkdir opt`”)

```

GNU nano 7.2 /opt/sonarqube/conf/sonar.properties
# IMPORTANT:
# - The embedded H2 database is used by default. It is recommended for tests but not for
#   production use. Supported databases are Oracle, PostgreSQL and Microsoft SQLServer.
# - Changes to database connection URL (sonar.jdbc.url) can affect SonarSource licensed products.

# User credentials.
# Permissions to create tables, indices and triggers must be granted to JDBC user.
# The schema must be created first.
#sonar.jdbc.username=
#sonar.jdbc.password=
sonar.jdbc.username=sonar
sonar.jdbc.password=sonar
sonar.jdbc.url=jdbc:postgresql://localhost/sonarqube
# Required project info
sonar.projectKey=Sonar_Python
sonar.sources=.
sonar.python.version=3.10

#----- Oracle 19c/21c
# The Oracle JDBC driver must be copied into the directory extensions/jdbc-driver/oracle/.
# Only the thin client is supported, and we recommend using the latest Oracle JDBC driver. See
# https://jira.sonarsource.com/browse/SONAR-9758 for more details.
# If you need to set the schema, please refer to http://jira.sonarsource.com/browse/SONAR-5000
#sonar.jdbc.url=jdbc:oracle:thin:@localhost:1521/XE
|
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo      M-C Copy

```

`sonar.jdbc.username=sonar`

`sonar.jdbc.password=sonar`

`sonar.jdbc.url=jdbc:postgresql://localhost/sonarqube`

`# Required project info`

`sonar.projectKey=Sonar_Python`

`sonar.sources=.`

`sonar.python.version=3.10`

Step: 5 Run the sonarqube

- `cd /opt/sonarqube/bin/linux-x86-64`
- `./sonar.sh start`
- `./sonar.sh status`

Step:6 Log in to sonarqube

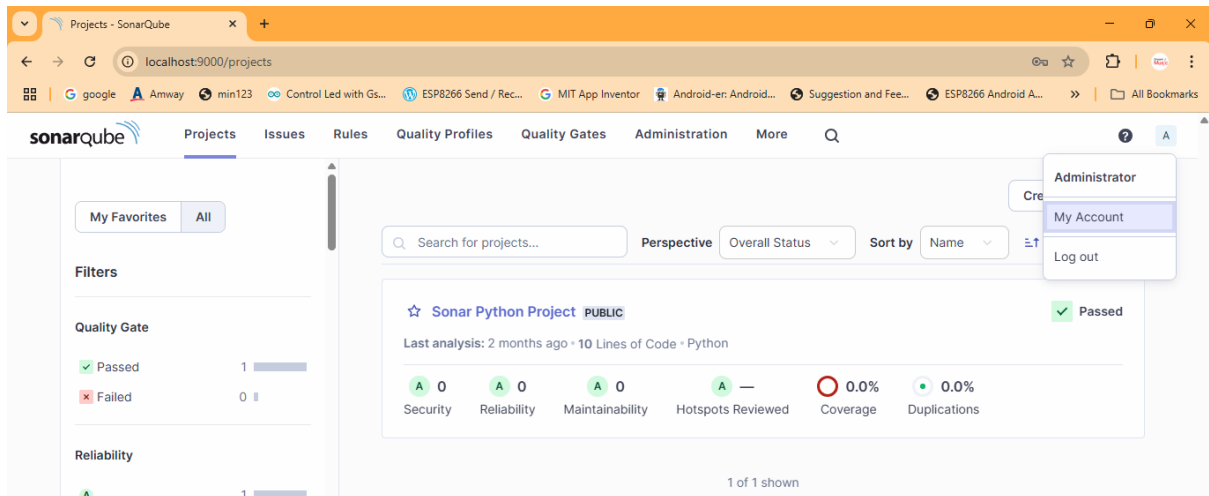
Goto> browser> localhost:9000

(wait for atleast 5-10 min, as it will take a time to start)

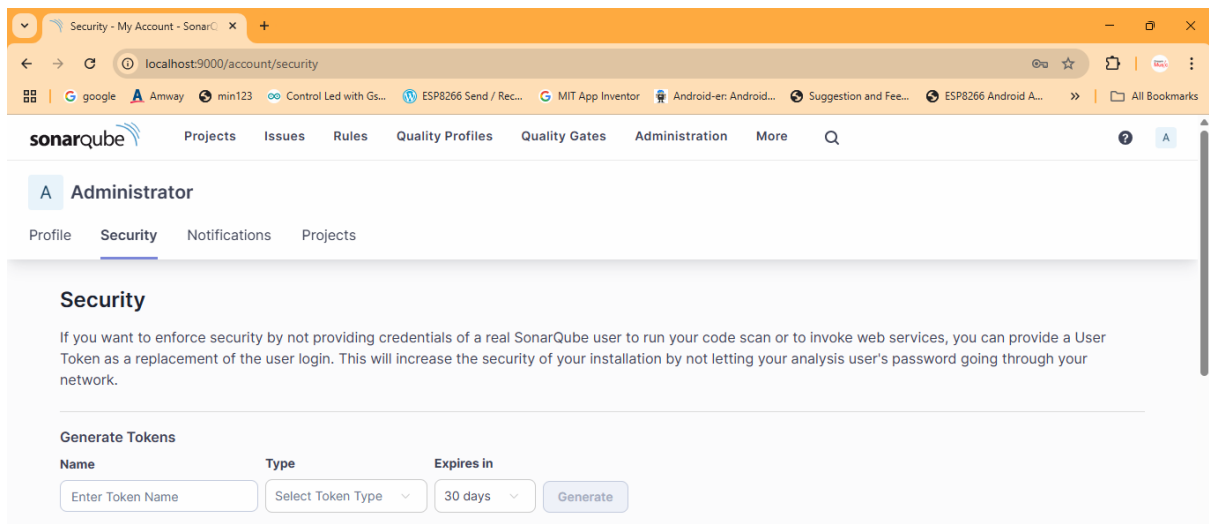
- Default;t username and password are
 - Username: admin
 - Password: admin

- Change the password on first Login

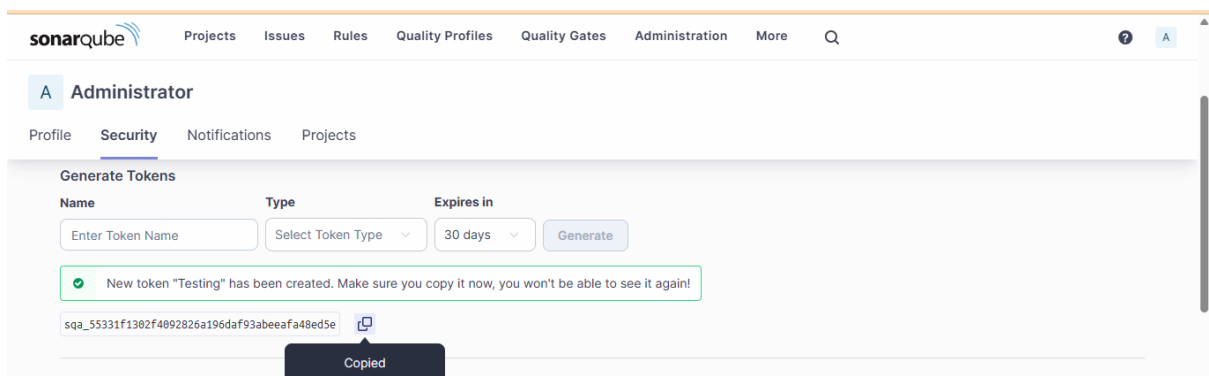
Now click on MyAccount



Click on security tab



Generate New Global Analysis Token



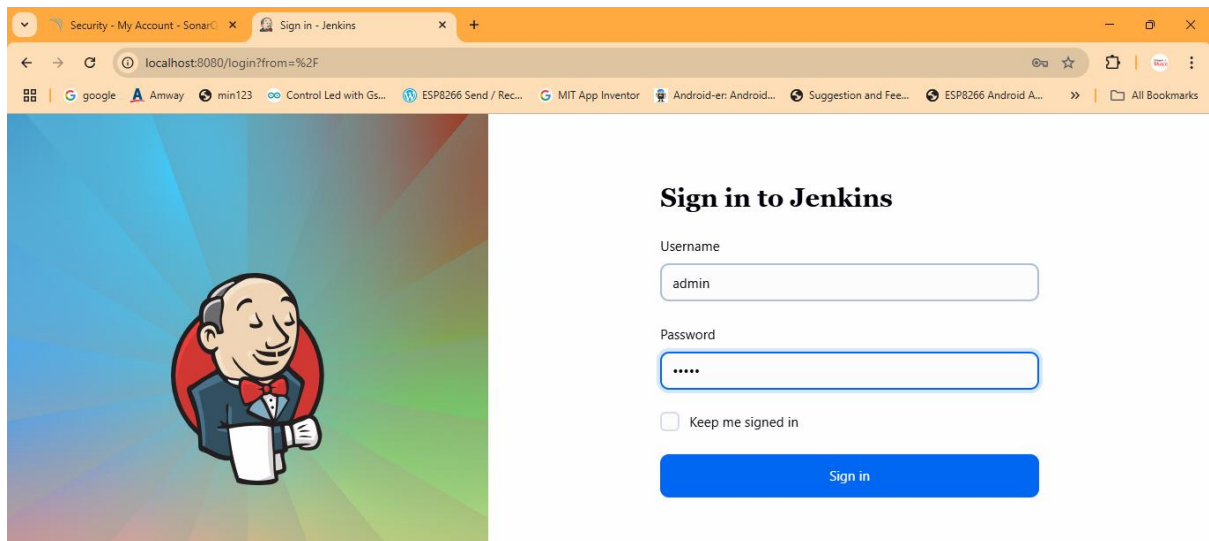
Copy it and save it somewhere securely

JENKINS SONARQUBE INTEGRATION

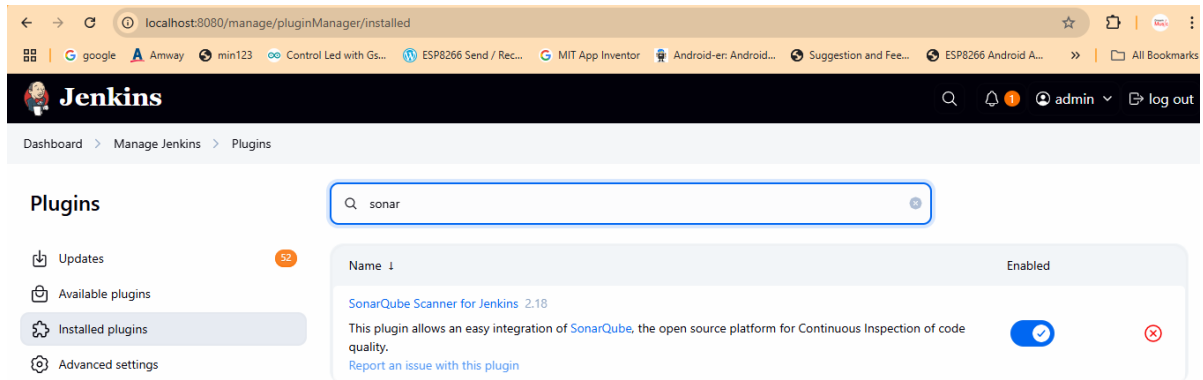
Step:7 start the Jenkins

```
nikunj@DESKTOP-M4S6FPT:/opt/sonarqube/bin/linux-x86-64$ sudo systemctl start jenkins
[sudo] password for nikunj:
nikunj@DESKTOP-M4S6FPT:/opt/sonarqube/bin/linux-x86-64$ |
```

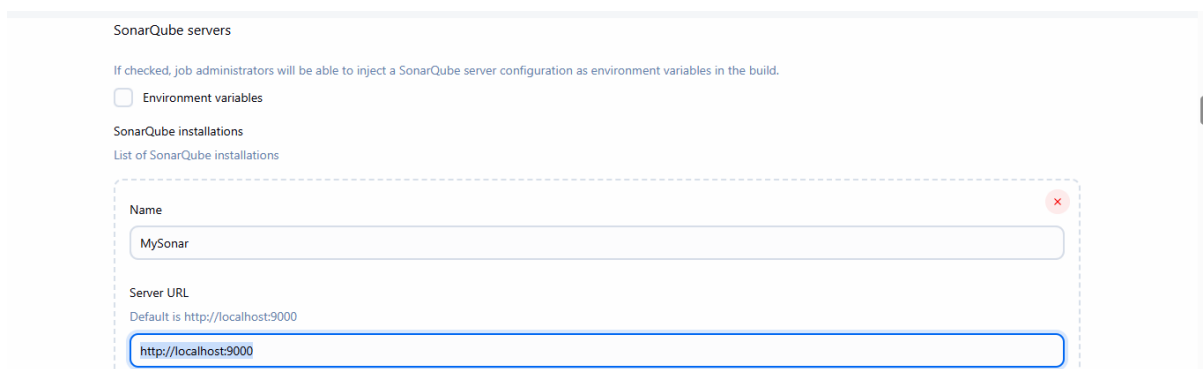
Step: 8 Open Jenkins and add sonarqube to it



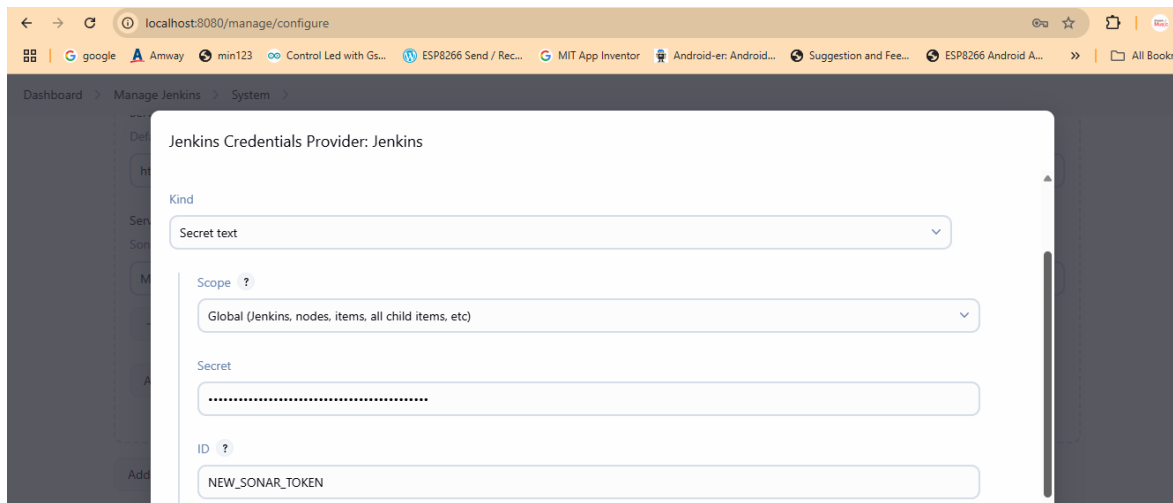
Goto> manage Jenkins> add sonarqube server for jenkins



Step:9 goto> manage Jenkins> tools(system)>

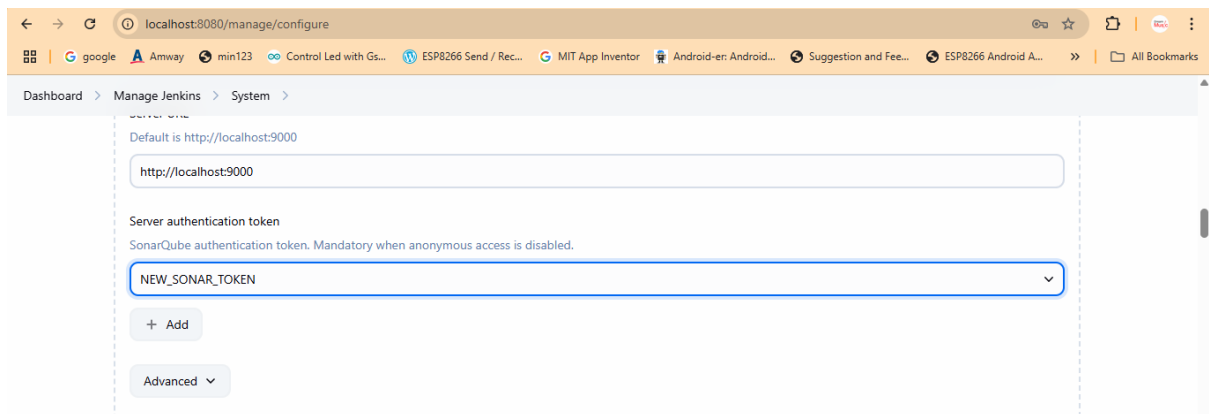


Then in Server authentication choose> Jenkins> add new token



The screenshot shows the 'Jenkins Credentials Provider: Jenkins' configuration window. The 'Kind' dropdown is set to 'Secret text'. The 'Scope' dropdown is set to 'Global (Jenkins, nodes, items, all child items, etc)'. The 'Secret' field is filled with a series of dots. The 'ID' field is set to 'NEW_SONAR_TOKEN'.

Now in server authentication dropdown > choose newly added token>



The screenshot shows the 'Jenkins System' configuration page. The 'Server URL' field is set to 'http://localhost:9000'. The 'Server authentication token' dropdown is set to 'NEW_SONAR_TOKEN'. There is an '+ Add' button and an 'Advanced' dropdown menu.

Then click on Save